

Manasvi Kushwah

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Executive Summary

final-year student with strong skills in Python, SQL, Machine Learning, and analytical modeling. Experienced in data analysis, signal processing, and ML pipeline development for real-world sensing applications. Interested in Data Analytics, Operations Research, and AI-driven decision systems, with hands-on experience in predictive modeling, feature engineering, and hardware-software integration.

Education

VIT Bhopal University	2022 – 2026
B.Tech in Electronics and communication engineering	
Specialization: AI and Cybernetics	Cgpa: 8.58/10

Technical Skills

Soft skills: Active listening, Communication, Team Management, Leadership
Languages: Python, C++, MATLAB, SQL, Embedded C, MATLAB
Data Analysis & ML: Pandas, NumPy, Scikit-learn, TensorFlow, Feature Engineering, EDA
Visualization & Tools: Matplotlib, Seaborn, MATLAB, Excel, LTspice, Tinkercad
Optimization & Modeling: Statistical Analysis, Predictive Modeling, Signal Processing
Developer Tools: Arduino IDE, Keil uVision, Linux, Git

Projects

ML Based Tuberculosis Detection using Cough Audio Python, TensorFlow, Scikit-Learn	August 2025 – Dec 2025
- Built an end-to-end data pipeline for preprocessing and classification of medical audio datasets - Performed feature extraction using MFCC and conducted exploratory data analysis on audio signals - Trained CNN, SVM, and ensemble models for predictive classification	
Low-Cost Spectrophotometer for Water Quality Analysis Python, MATLAB, Signal Processing	Dec 2024 – May 2025
- Engineered a custom spectrophotometer using a webcam, diffraction grating, and photoresistor, focusing on hardware integration and optical calibration. - Developed signal processing algorithms in MATLAB to analyze spectral data, identifying contaminants with a calibrated sensor accuracy of $\pm 5\%$ error.	
Sparse RBM for Blood Cancer Classification Python, TensorFlow	Nov 2024 – Dec 2024
- Developed Sparse RBM for high-dimensional gene expression data. - Achieved 82% classification accuracy for AML & ALL types - Demonstrated strong understanding of statistical modeling and data-driven classification	

Certifications

Applied Machine Learning in Python - University of Michigan | Coursera
MATLAB Onramp, Signal Processing Onramp, & Computer Vision - MathWorks Academy
VLSI Design Training - Maven Silicon

Experience

Maven Silicon Internship in Embedded System Design (Feb-May 2025)
Demonstrated applied proficiency with Arduino UNO Rev3, Raspberry Pi 3, and Node MCU boards effectively integrated diverse sensors to gain knowledge of 3 Serial Communication protocols

Extracurriculars

- Smart India Hackathon (SIH) 2023 Participant : Shortlisted for the internal round for proposing an innovative robotics Solution (Hexabot)
- Participated in Jatayu Virtusa Ideathon as a Team Lead, focused on developing industry-oriented solutions, contributing to problem analysis, idea presentation, and collaborative strategy building.